Curriculum Vitae

Ying Kung

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EDUCATION

National Taiwan University (NTU) Taiwan
M.S. in applied mechanics Aug 2023 — Jun 2025
GPA: 3.67/4.3
Relevant courses: Applications of Deep Learning in Science and Engineering; Principles and Experiments of Mechatronic Systems; Stochastic Control; Electronics Laboratory

National Yang Ming Chiao Tung University (NYCU) Taiwan
B.S. in Civil Engineering Aug 2019 — Jun 2023

RESEARCH INTERESTS

hybrid physics–ML frameworks for process modeling; inverse methods for behavior reconstruction; modeling and algorithms for estimation, control, and health diagnostics; energy management of auto mated, connected, and electrified vehicles; real-time system identification and state estimation; data-driven prediction and optimization.

PUBLICATIONS

- **Ying Kung**, Yu-Hong Zhang, Chi-Jyun Ko, & Kuo-Ching Chen (2025). Predicting the onset of lithium plating by the full-cell voltage: A pseudo-P curve approach. *Energy Storage Materials* (accepted).

CONFERENCE PRESENTATIONS

- **Ying Kung**. Onset prediction of lithium plating from full-cell voltage and its application to charging strategies. Presented at the *35th Annual Conference on Combustion and Energy*, National Tsing Hua University, Hsinchu, Taiwan, May 2025.
- **Ying Kung**. Frequency-specific modeling for battery parameter identification under battery aging. presentated at *The 2025 International Conference on Green Electrochemical Technologies & 2025 Annual Meeting of Electrochemical Society of Taiwan (ICGET-Tw)*, National Cheng Kung University, Tainan, Taiwan, Nov 2025.

RESEARCH PROJECTS

Segmented System Identification for Nonlinear Dynamical Models
Graduate Research | Advisor: Prof. Kuo-Ching Chen NTUIAM, May 2025 — Present

- Developed a segmented system-identification framework integrating lightweight Transformer-based regressors with physics-informed structural constraints to achieve real-time parameter estimation of nonlinear dynamical systems.
- Applied two-stage Sobol global sensitivity analysis to decompose system responses into frequency-specific subproblems, enabling efficient domain decomposition and improved identifiability.
- Achieved sub-second inference under both nominal and degraded conditions, supporting potential deployment in robotic state estimation and adaptive control.
- Reduced computational load by five orders of magnitude relative to full-spectrum inversion, demonstrating suitability for onboard or resource-limited systems.

Safety Boundary Detection and Control-Oriented State Prediction
Graduate Research | Advisor: Prof. Kuo-Ching Chen NTUIAM, Nov 2024 — Apr 2025

- Formulated a physics-based criterion for detecting critical-state transitions in nonlinear dynamical systems using only input–output measurements.
- Demonstrated that the criterion functions as a reachability-based safety boundary, enabling anticipatory detection of unsafe transitions under real-time constraints.
- Validated the method through large-scale simulations and controlled experiments, showing reliable detectability without auxiliary sensing.
- Designed multi-stage, constraint-aware input strategies that maintain performance while avoiding unsafe states—analogous to safe planning under model uncertainty in robotic systems.

Concrete Boat Design and Fabrication Project
Undergraduate Capstone Project | Advisor: Assoc. Prof. Hsin-yu Shan NYCUCV, Jan 2023 — Jun 2023

- Designed and built a functional concrete boat; conducted buoyancy and structural analysis, created CAD models, and fabricated molds via 3D printing and machining.

AWARDS & SCHOLARSHIPS

- Honorable Mention, Student Paper Competition, 35th National Conference on Combustion and Energy May 2025
- Dean's Award, College of Engineering, National Taiwan University Aug 2025

SKILLS

- **Modeling & Control:** nonlinear dynamics; system identification; estimation; real-time control; optimization
- **Programming & Simulation:** Python, MATLAB, PyTorch, COMSOL
- **Battery Modeling (domain expertise):** PyBaMM, electrochemical modeling, EIS analysis
- **Documentation & Tools:** LaTeX, Git